Height Quotients of the Riemann Zero Divisor

Central-circle limits and invisible sparse perturbations

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Abstract

We study fixed-height identifications of the centered critical strip, with ordinary or half-twisted gluing. Classical equidistribution and zero-density theorems imply an unconditional central-circle Haar limit for the normalized zero measures on both resulting compact surfaces. We give a quantitative Lipschitz bound, prove infinite raw mass near every central-circle point, and exclude eventual periods of the unweighted zero support. A packing argument also gives a logarithmic lower bound for the number of residues of simple critical-line zeros. An explicit even entire product then inserts infinitely many sparse off-critical zeros while preserving these probability limits, the proved smooth prime-resonant secondary statistics, and the usual zero-counting asymptotic class. After normalization it also preserves the zeta functional equation and pole residue, but fails an ordinary Dirichlet series condition. The paper is a synthesis and obstruction study using classical arithmetic inputs; it makes no priority claim for those inputs or claim of a proof or counterexample to the Riemann hypothesis.

1 Question, contribution, and arithmetic inputs

We use Riemann's entire function, with removable values understood,

$$\xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \qquad \xi(1-s) = \xi(s).$$

This is the normalization of [1, 25.4.3–25.4.4]; it also satisfies $\xi(\bar{s}) = \overline{\xi(s)}$. Center the critical strip by writing

$$\rho = \tfrac{1}{2} + x_\rho + i\gamma, \qquad -\tfrac{1}{2} < x_\rho < \tfrac{1}{2}.$$

A drawing that identifies heights modulo a fixed length always produces a quotient of the ambient strip. It need not produce an invariant quotient of its marked zero divisor. A second distinction is just as important: the raw pushforward and its normalized finite-height probability measures have very different behavior. We make these distinctions precise for both a cylinder and a Möbius band, and ask which properties of the zeros their limiting observations can determine.

The arithmetic inputs are prior results. Our contribution is the connected quotient formulation, its quantitative collapse estimate, and an explicit infinite sparse perturbation testing the information retained by these observations. The support and packing results below are elementary corollaries of the cited inputs; the product construction uses standard infinite-product analysis. We do not assert that this combination has priority over every possible earlier formulation.

Let m_ρ be the order of a nontrivial zero and put

$$N(H) = \sum_{0<\operatorname{Im}\rho\le H} m_\rho.$$

Every sum over zeros in this paper includes m_ρ unless distinct support or simple zeros are explicitly specified. We use the following unconditional inputs, with their exact scopes recorded here.

1. The Riemann–von Mangoldt asymptotic is

$$N(H) = \frac{H}{2\pi}\log\frac{H}{2\pi} - \frac{H}{2\pi} + O(\log H). \tag{1}$$

   The count is finite in every bounded-height slab, since Riemann's entire function ξ is nonzero and its zeros are precisely the nontrivial zeta zeros, with orders.

2. For every fixed $\alpha \neq 0$, the fractional parts $\{\alpha\gamma\}$ of all nontrivial zero ordinates are uniformly distributed modulo one. Ford–Zaharescu [3, Introduction] state Hlawka's unconditional theorem and record Fujii's unconditional star-discrepancy bound

$$D^*_\alpha(H) = O_\alpha\left(\frac{\log\log H}{\log H}\right). \tag{2}$$

   The original Hlawka and Fujii articles are identified in [3]; our direct source for these precisely stated consequences is that research paper.

3. If $N(\sigma, H)$ counts zeros with multiplicity, with $|\gamma| \leq H$ and $\operatorname{Re}\rho \geq \sigma$, Selberg's bound as stated in [3, Lemma 2] is, uniformly for $1/2 \leq \sigma \leq 1$,

$$N(\sigma, H) \ll H^{1-(\sigma-1/2)/4}\log H. \tag{3}$$

4. Bui–Conrey–Young [2, Theorem 1.1] prove

$$\liminf_{H\to\infty} \frac{N_0^*(H)}{N(H)} \geq 0.4058, \tag{4}$$

   where $N_0^*(H)$ counts *simple critical-line* zeros of positive height. We use the conservative constant $c = 0.4$.

The secondary smooth-test theorem of [3] is stated in Section 5. No interval version of that theorem is assumed. All inputs concern actual zeta zeros. They do not assert equidistribution of the critical-line-only subsequence.

We consistently use $0 < \gamma \leq H$. When a cited count uses a strict or averaged endpoint, squeeze it between nonordinate heights within distance one. The main term in (1) changes by $O(\log H)$ on such an interval; the same asymptotic class follows. For bounded smooth tests, any endpoint contribution to a $1/H$-scaled statistic is $O(\log H/H)$. In particular, changing the BCY endpoint count costs $O(\log H) = o(N(H))$, so its limiting proportion is unaffected.

2 Dense winding and support-period obstructions

Let S be the unweighted zero support in centered coordinates. For $L > 0$ write $\mathbb{T}_L = \mathbb{R}/L\mathbb{Z}$, with arc length $|I|$ and normalized Haar probability m_L. An eventual support period L means that, above some height H_0, membership is equivalent under $(x,t) \mapsto (x, t+L)$ whenever both heights are in that tail. Mere forward inclusion is a different hypothesis.

Theorem 2.1. *For every $L > 0$, every nonempty open arc of $\mathbb{T}_L$ contains infinitely many distinct residues of positive zero ordinates. The complete unweighted support has no full or eventual positive vertical period. It also has no full or eventual glide period $(x,t) \mapsto (-x, t+L)$.*

Proof. Uniform distribution gives, for every arc I,

$$\sum_{\substack{0<\gamma\le H\\ [\gamma]_L\in I}} m_\rho = \left(\frac{|I|}{L}+o(1)\right)N(H). \tag{5}$$

In particular every open arc is met. If an open arc contained only finitely many distinct projected ordinates, choose a smaller nonempty open subarc avoiding them and apply (5) again. This proves the asserted infinitude, without discarding multiplicities from the equidistribution input.

Under eventual period equivalence, every sufficiently high zero can be translated downward into a slab of height L above H_0. There are only finitely many support points in that slab: embed it in a compact rectangle of the closed strip and use local finiteness of the entire ξ divisor. The finite positive-height prefix also has only finitely many points. All residues would thus belong to a finite set, contradicting the first assertion. Squaring an eventual glide gives an eventual vertical period $2L$, which is already excluded. □

For comparison, the simpler divisor argument only uses (1): periodicity of the *multiplicity divisor* gives the same finite mass in each half-open height cell, hence $N(H) = O(H)$, a contradiction. That argument alone does not exclude periodic support with varying multiplicities. Theorem 2.1 supplies the missing equidistribution input.

Remark 2.2. There is no assertion here that individual gaps never recur, that approximate patterns cannot recur, or that an unrelated dynamical model cannot have periodic orbits. The obstruction concerns exact fixed-height repetition of the specified support or divisor.

3 Cylinder and Möbius probability limits

Fix a closed width $A \ge 1/2$, a height length $L > 0$, and $e \in \{1, -1\}$. On $B_A = [-A, A] \times \mathbb{R}$ let $d_e(x, t) = (ex, t + L)$, and set $Q_e = B_A/\langle d_e\rangle$. The case $e = 1$ is a closed cylinder; $e = -1$ is a closed Möbius band. Denote the quotient map by q_e. The equivalence relation means the entire integer deck orbit, not just one identification.

Lemma 3.1. *The quotient topology is given by the metric*

$$d_{Q_e}([x, t], [y, u]) = \inf_{n\in\mathbb{Z}} \sqrt{(x - e^n y)^2 + (t - u - nL)^2}. \tag{6}$$

The space is compact. Its central embedding $j_e : \mathbb{T}_L \to Q_e$, $j_e([t]) = [0, t]$, is isometric. The retraction $r_e([x, t]) = [0, t]$ is well defined and satisfies

$$d_{Q_e}([x, t], r_e([x, t])) = |x|. \tag{7}$$

Proof. The deck action is by Euclidean isometries, so the infimum is independent of representatives and satisfies the triangle inequality. Only finitely many n can keep $|t - u - nL|$ bounded. Thus distance zero forces one deck translate to coincide with the other representative. Also the inverse image of each quotient ball is a union of Euclidean balls in the relative strip. Conversely a deck-saturated open set contains a ball about every representative, so its image is metric-open. This verifies the topology claim. The fundamental closed rectangle $[-A, A] \times [0, L]$ maps onto Q_e, proving compactness. With both transverse coordinates zero, (6) reduces to the usual circle distance. The formula for r_e respects the deck relation, and the vertical circle distance is bounded by each term in (6); hence the retraction is continuous. Setting $y = 0, u = t$ proves (7). □

Define probability measures, for $N(H) > 0$,

$$\nu_{e,H} = \frac{1}{N(H)} \sum_{0<\gamma\le H} m_\rho \, \delta_{q_e(x_\rho,\gamma)}, \qquad \lambda_{L,H} = \frac{1}{N(H)} \sum_{0<\gamma\le H} m_\rho \, \delta_{\{\gamma/L\}}. \tag{8}$$

Write $\mu_e = (j_e)_* m_L$. On a compact metric space define

$$d_{\mathrm{Lip}}(\nu,\mu) = \sup_{\mathrm{Lip}(f)\le 1} \left| \int f \, \mathrm{d}\nu - \int f \, \mathrm{d}\mu \right|,$$

where tests are real-valued. Constants cancel because both masses are one.

Theorem 3.2. *For either deck sign, $\nu_{e,H}$ converges weakly to μ_e, unconditionally. More precisely,*

$$\begin{aligned} d_{\mathrm{Lip}}(\nu_{e,H},\mu_e) &\le \frac{1}{N(H)} \sum_{0<\gamma\le H} m_\rho |x_\rho| + L D^*_{1/L}(H) \\ &= O_L\!\left(\frac{\log\log H}{\log H}\right). \end{aligned} \tag{9}$$

The weak-limit assertion only requires Hlawka equidistribution and (3); the displayed rate also uses (2).

Proof. Critical-line reflection preserves multiplicity at each positive height. Layer cake and (3), with its uniform constant C, give

$$\begin{aligned} \sum_{0<\gamma\le H} m_\rho |x_\rho| &= \int_0^{1/2} \sum_{\substack{0<\gamma\le H \\ |x_\rho|>v}} m_\rho \, \mathrm{d}v \\ &\le 2CH\log H \int_0^{1/2} e^{-v\log H/4} \, \mathrm{d}v \le 8CH. \end{aligned}$$

The two-sided convention in the density input only enlarges this bound. By (7), replacing $f(q_e(x_\rho,\gamma))$ by $f(j_e([\gamma]_L))$ costs at most the first term of (9). The function $g(u) = f(j_e([Lu]))$ on $[0,1]$ is periodic and L-Lipschitz. It is absolutely continuous. Stieltjes integration by parts, with $F_H(u) = \lambda_{L,H}([0,u))$, gives

$$\left| \int g \, \mathrm{d}(\lambda_{L,H} - \, \mathrm{d}u) \right| \le \sup_u |F_H(u) - u| \int_0^1 |g'(u)| \, \mathrm{d}u \le L D^*_{1/L}(H).$$

Periodic endpoints cancel; left and right CDF conventions differ only at atoms and have the same integral against g'. This proves the inequality. Now use (1) and (2) for the rate.

For completeness, weak convergence for every continuous complex-valued test does not require a transport duality theorem. Uniform continuity on Q_e makes its retraction error uniformly small on $|x| < \epsilon$. The exceptional proportion is $O(H^{-\epsilon/4})$ by (3) and (1). The restriction to the central circle is handled by Hlawka. Letting ϵ decrease after taking the limit proves the claim. □

Proposition 3.3. *The raw positive-height pushforward $\mathcal{D}_e = \sum_{\gamma>0} m_\rho \delta_{q_e(x_\rho,\gamma)}$ has infinite mass in every neighborhood of every central-circle point. It is not locally finite and is not a Radon measure.*

Proof. Such a neighborhood contains the image of a narrow transverse strip $|x| < \delta$ above a nonempty open height arc I. The metric formula justifies this statement even across the twisted seam. Through height H its mass is at least the count in (5) minus the count with $|x| \ge \delta$. The former is $(|I|/L + o(1))N(H)$; the latter is $o(N(H))$. Their difference diverges. A Radon measure is finite on compact sets, whereas Q_e is compact and already has infinite mass. □

No analogous conclusion at every off-center point is asserted: RH would leave an off-center neighborhood with no zeros. Theorem 3.2 does not say that each zero is on the central circle, only that the weighted exceptional proportion vanishes. It therefore does not prove RH.

4 Critical-line residue complexity

Let Γ_+ be the distinct positive critical-line ordinates, and let $R_L(H)$ count residues modulo L represented by *simple* critical-line zeros through height H.

Theorem 4.1. *For every fixed $L > 0$,*

$$R_L(H) \geq \frac{N_0^*(H)}{\lfloor H/L \rfloor + 1}, \qquad \liminf_{H\to\infty} \frac{R_L(H)}{\log H} \geq \frac{0.4L}{2\pi}. \tag{10}$$

In particular Γ_+ has no eventual support period.

Proof. Simple critical-line zeros have distinct ordinates. A residue class modulo L contains at most $\lfloor H/L \rfloor + 1$ positive heights through H, proving the first inequality. By (4), eventually $N_0^*(H) \geq 0.4N(H)$. Divide by $\log H$ and use $N(H)/(H \log H) \to 1/(2\pi)$ and $(\lfloor H/L \rfloor + 1)/H \to 1/L$. An eventual period would leave only finitely many residues, contradicting their unbounded number. □

Proposition 4.2. *Let K_L be the closure in $\mathbb{T}_L$ of the projected positive critical-line ordinates, or of any fixed tail of them. Then $m_L(K_L) \geq 0.4$.*

Proof. The all-zero height probability assigns K_L at least the proportion of simple critical-line zeros, apart from a finite prefix. Its limiting lower mass is at least 0.4 by (4). Since K_L is closed, weak convergence to Haar and the closed-set inequality imply that its Haar mass is at least that number, explicitly

$$0.4 \leq \liminf_{H\to\infty} \lambda_{L,H}(K_L) \leq \limsup_{H\to\infty} \lambda_{L,H}(K_L) \leq m_L(K_L),$$

identifying the unit circle with $\mathbb{T}_L$ by $u \mapsto [Lu]_L$. □

Neither this positive-measure statement nor (10) proves that the critical-line-only sequence is equidistributed or dense. Those claims do not follow merely by selecting a positive-proportion subsequence.

5 Stability of the observations

Lemma 5.1. *Let D be a finite positive atomic measure of mass $N > 0$, and E another of mass $M \geq 0$. For every bounded measurable test h on their common target space,*

$$\left| \frac{\int h \,\mathrm{d}(D+E)}{N+M} - \frac{\int h \,\mathrm{d}D}{N} \right| \leq \frac{2\|h\|_\infty M}{N+M}. \tag{11}$$

For a reference probability m and $H > 0$,

$$\left| \frac{\int h \,\mathrm{d}E - M \int h \,\mathrm{d}m}{H} \right| \leq \frac{2\|h\|_\infty M}{H}. \tag{12}$$

Proof. Put $A = \int h \,\mathrm{d}D$, $B = \int h \,\mathrm{d}E$, and $I = \int h \,\mathrm{d}m$. The numerator in (11), after putting it over the denominator $N(N+M)$, is $NB - MA$. Use $|A| \leq \|h\|_\infty N$ and $|B| \leq \|h\|_\infty M$. The other inequality follows from $|B - MI| \leq |B| + M|I|$. □

For a zero divisor D define the smooth secondary functional on $\mathbb{T}_1$ by

$$\mathcal{S}^D_{\alpha,H}(h) = \frac{1}{H}\left(\sum_{0<\gamma\le H} m_\rho h(\{\alpha\gamma\}) - N_D(H)\int_0^1 h(u)\,\mathrm{d}u\right). \tag{13}$$

Ford–Zaharescu [3, Theorem 1 and Corollary 2] prove for $h \in C^2(\mathbb{T}_1)$ that this converges to $\int h g_\alpha$ for zeta. For $\alpha > 0$ the density is zero unless $\alpha = r\log p/(2\pi q)$ for a prime p and coprime positive integers r, q. In the latter case

$$g_\alpha(u) = -\frac{\log p}{\pi}\operatorname{Re}\sum_{k\ge1} p^{-rk/2}e^{-2\pi iqku}. \tag{14}$$

The series is absolutely convergent. These are smooth-test secondary limits, not weak convergence against every continuous test of the unbounded signed measures, and not an established interval-indicator theorem.

6 An infinite off-critical perturbation

Fix $0 < \eta < 1/2$ and put $z_n = \eta + i2^n$ for $n \ge 1$. Define

$$P(w) = \prod_{n\ge1} A_n(w), \qquad A_n(w) = (1 - w^2/z_n^2)(1 - w^2/\bar z_n^2). \tag{15}$$

The exponential heights are chosen to retain an $O(\log H)$ counting perturbation; no arithmetic significance of the base two is claimed.

Theorem 6.1. *The product is an even real entire function with simple zeros exactly $\{\pm\eta \pm i2^n : n \ge 1\}$. It is positive on the real and imaginary axes. For $|w| \le R$,*

$$\sum_{n>N} |A_n(w) - 1| \le B_N(R) = \frac{2R^2}{3}4^{-N} + \frac{R^4}{15}16^{-N}. \tag{16}$$

If $B_N(R) < 1$, its tail factor satisfies

$$\left|\prod_{n>N} A_n(w) - 1\right| \le \frac{B_N(R)}{1 - B_N(R)}. \tag{17}$$

Furthermore $\log\max_{|w|\le R}|P(w)| = O((\log(R+2))^2)$.

Proof. Because $|z_n| \ge 2^n$, expansion of the paired factor gives

$$|A_n(w) - 1| \le 2R^2 4^{-n} + R^4 16^{-n}.$$

Its compact-uniform summability proves normal convergence of the product, hence entire holomorphy. At a point outside the prescribed zeros, the tail has a nonzero limit by the summable-deviation infinite-product criterion; equivalently take a sufficiently small tail and use its convergent logarithm. Near a prescribed zero, all the other factors form a nonvanishing analytic function. The four roots of each factor are distinct and different factors have distinct heights. This proves the exact simple divisor.

Evenness and conjugation symmetry hold factorwise. For real u, v,

$$A_n(u) = |1 - u^2/z_n^2|^2, \qquad A_n(iv) = |1 + v^2/z_n^2|^2.$$

They are positive because z_n^2 is nonreal. Their infinite product cannot vanish there, by the criterion just proved, so the limit is positive.

Summing the two geometric tails gives (16). For a finite set of tail factors put $u_n = |A_n(w) - 1|$. Expanding the product and taking absolute values bounds its difference from one by the sum of its elementary symmetric polynomials in the u_n. The degree-k sum is at most $(\sum u_n)^k$, so the difference is at most $\sum_{k\geq 1} B_N^k = B_N/(1 - B_N)$. Pass to the product limit to obtain (17). This is a bound on the tail factor, even where the finite prefix is zero; no division by that prefix is required.

Finally $|A_n(w)| \leq (1 + R^2 4^{-n})^2$. Split the sum of its logarithms at $J = \lceil \log_2 \max(2, R) \rceil$. The first J terms are individually $O(\log(R + 2))$, and the remaining logarithms have bounded sum using $\log(1 + v) \leq v$. This proves the maximum-modulus bound. In particular P has entire-function order zero. □

Theorem 6.2. *Let $X(s) = \xi(s)P(s - 1/2)$. It has infinitely many prescribed off-critical zeros in the open strip, satisfies $X(1 - s) = X(s)$ and the real conjugation symmetry, and retains every critical-line zero with its original order. Its real critical-line restriction has the same sign away from zeros. For $H \geq 2$,*

$$N_X(H) = N(H) + M(H), \qquad M(H) = 2\lfloor \log_2 H \rfloor. \tag{18}$$

It has the same cylinder and Möbius probability limits as zeta, and

$$\mathcal{S}^X_{\alpha,H}(h) - \mathcal{S}^\xi_{\alpha,H}(h) \to 0 \tag{19}$$

for every fixed α and bounded measurable test h. Consequently all the proved smooth limits (14) persist.

Proof. Divisors of nonzero holomorphic products add, including at accidental coincidences with xi zeros. Theorem 6.1 inserts exactly two positive-height roots for each $2^n \leq H$, proving (18) with the closed endpoint convention. The symmetry and sign assertions follow from the even real structure and the positive imaginary-axis values of P. Exact function values on the critical line are not preserved. Since $M(H) = O(\log H) = o(N(H))$, (11) transfers the quotient probability limits. Inequality (12) bounds the difference in (19) by $2\|h\|_\infty M(H)/H = O(\|h\|_\infty \log H/H)$. This transfers a limit whenever it exists; it does not establish an unproved interval limit for either function. □

The Riemann–von Mangoldt main terms and its $O(\log H)$ remainder class are unchanged. The density bound (3) is unchanged as a big-O class too: the additional two-sided count is $O(\log H)$, absorbed uniformly since $1 - (\sigma - 1/2)/4 \geq 7/8$ on the stated range. The quotient rate is likewise unchanged in order by (11).

Proposition 6.3. *For every fixed $x > 1$, the leading term and error class in Landau's formula*

$$\sum_{0<\gamma\leq H} m_\rho x^\rho = -\frac{H}{2\pi}\Lambda(x) + O_x(\log H)$$

are preserved by X. Here $\Lambda(x)$ is zero unless x is an integral prime power. The error class of the uniform version in [3, Lemma 1] also absorbs the perturbation for $x > 1, H \geq 2$.

Proof. Every added zero has real part between zero and one, so its contribution has modulus at most x. The total difference is at most $xM(H) = O(x \log H)$. For fixed x this gives the first claim. For varying x, the cited uniform error is $O(x \log^2(2xH) + \log(2H)/\log x)$. Its first component absorbs this bound. The nearest-prime-power main term is not changed. □

This assertion concerns asymptotic classes with unspecified constants. It does not preserve an exact Weil explicit formula or an Euler product. An exact added-divisor term remains present and can be detected by suitable finite-height tests.

7 The arithmetic boundary: an ordinary Dirichlet series

The preceding construction can also match the pole normalization. Put

$$\widetilde{\zeta}(s) = \zeta(s)\frac{P(s-1/2)}{P(1/2)}. \tag{20}$$

The denominator is a strictly positive constant.

Proposition 7.1. *The meromorphic function $\widetilde{\zeta}$ has the same gamma-factor functional equation as zeta, the same pole at one with residue one, and the same trivial-zero multiplicities. Nevertheless it is not representable by any ordinary Dirichlet series absolutely convergent on a right half-plane.*

Proof. The multiplier is entire, nonzero on the real axis, and invariant under $s \mapsto 1-s$. Multiplying zeta's meromorphic functional equation therefore gives the same equation, with its usual gamma factor. Its value at $s = 1$ is one, proving the residue statement. It is nonzero at each negative even integer, so it retains every trivial-zero multiplicity.

For real u, direct expansion gives

$$A_n(u) = 1 + \frac{2(4^n - \eta^2)}{(\eta^2 + 4^n)^2}u^2 + \frac{u^4}{(\eta^2 + 4^n)^2} \geq 1.$$

Thus $P(u) \geq A_1(u) \to \infty$ as $u \to \infty$. Since $\zeta(\sigma) \to 1$ on the positive real axis, (20) tends to infinity there. In contrast, if $F(s) = \sum_{n\geq 1} a_n n^{-s}$ converges absolutely at a real σ_0, dominated convergence gives $F(\sigma) \to a_1$ as $\sigma \to \infty$: for $\sigma \geq \sigma_0$, the summands are dominated by the summable $|a_n|n^{-\sigma_0}$. This contradiction proves the claim. □

The role of an ordinary Dirichlet series is consonant with Hamburger's converse theorem, whose precise classical formulation is recalled in [5, Introduction]: if two ordinary Dirichlet series f, g converge absolutely for $\operatorname{Re} s > 1$, if $(s-1)f(s)$ and $(s-1)g(s)$ are entire of finite order, and if

$$\pi^{-s/2}\Gamma(s/2)f(s) = \pi^{-(1-s)/2}\Gamma((1-s)/2)g(1-s),$$

then $f = g = c\zeta$. We do not invoke that theorem to prove the proposition; the positive-real-axis obstruction suffices. Related non-Dirichlet functions with Riemann's functional equation are already studied, for example in [6]. Our claim is the specific sparse-stability package, not the general discovery that functional equations alone fail to characterize zeta or imply RH.

8 Verification, interpretation, and open problems

The exact verifiers supplied with this draft check the independent geometric tails in (16), finite complex paired-factor and tail deviations, prescribed roots, axis-positivity samples, dyadic count endpoints, and the fail-closed assertion guards. Such finite arithmetic does not certify holomorphy of an infinite product, its zero divisor, or a zeta theorem; those are the written arguments above. The real-test algebra in Lemma 5.1 is also kernel-checked in the companion `EmpiricalStability.lean` against Lean 4.33.1 and a pinned Mathlib checkout. Its reported logical axioms are `propext`, `Classical.choice`, and `Quot.sound`. No claim is made of a Lean proof of the analytic input theorems or of the whole manuscript. Earlier coordinate and zeta-support modules retain their separate, limited coverage.

The main conclusions distinguish three operations. An ambient height quotient always exists. Descent of the original locally finite divisor would require actual deck compatibility, which the support theorem excludes. Normalized finite-height measures do converge, but their central-circle limit and even the stated smooth secondary signatures tolerate an infinite sparse

off-critical divisor. The half-twist changes the ambient topology without repairing these information losses.

Several questions remain meaningful. Can one obtain stronger information about the projected critical-line-only subsequence without assuming its equidistribution? Which *exact* arithmetic identities, rather than the asymptotic observations listed here, distinguish the zero divisor of zeta from the sparse perturbations? How much finer a truncation-dependent test family is needed to detect a prescribed sparse quartet? The interval secondary asymptotics conjectured in [3, 4] are not settled here. No result in this paper proves or refutes RH.

References

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